Assignment 1

Dataset

https://drive.google.com/file/d/1IODZF8at54psKhh3bEMyU6sIk74_7nNm/view?usp=drive_link

This dataset contains daily sales data from a supermarket, including product categories, branch, customer type, gender, and total sales.

Tasks

1. Data Loading and Inspection

- Load the dataset into a Pandas DataFrame.
- Display the first 10 rows and inspect the basic structure.
- Report the number of rows and columns.
- Identify data types and missing values.

2. Data Cleaning and Preparation

- Handle any missing values appropriately.
- Convert the Date column to a datetime type.
- Create a new column **TotalSales** which represents the overall purchase amount if not already present.

3. Exploratory Data Analysis

- Identify the top 3 product lines contributing the most to total sales.
- Determine which branch has the highest average sales.
- Find the distribution of customer types (Member vs Normal) and identify which group spends more on average.

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4. Visualizations

- Plot a bar chart showing total sales by product line.
- Create a box plot to compare sales distribution across branches.
- Draw a histogram for the **TotalSales** column.
- Use Seaborn pairplot for numerical attributes to identify relationships.
- Plot a correlation heatmap for all numerical features.

5. Analytical Insight Questions

- Which product line generates consistent sales regardless of branch?
- Does gender have a significant influence on average sales value?
- Suggest one insight that could support a business decision.

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